MICHAEL GALLEGOS Multi-Disciplinary Engineer Colorado Springs, CO Phone: (719) 673-5372 Email: gallegosmike1997@gmail.com GitHub: github.com/gallegosmike1997 Portfolio: https://michael-gportfolio.vercel.app/ PROFESSIONAL SUMMARY Multi-disciplinary engineer with hands-on experience across software development, embedded systems, robotics, additive manufacturing, electrical engineering, AI integration, and full-stack application design. Proven ability to design, build, program, and deploy complex systems end-to-end. CORE COMPETENCIES • Software Engineering: Python, Rust, JavaScript, Android Studio, VS Code • AI & ML Integration: ONNX models, TTS pipelines, model routing • Embedded Systems: STM32, ST-Link, serial debugging, firmware flashing • Electrical Engineering: Circuit design, soldering, sensor integration • Mechanical & CAD: 3D CAD modeling, KiCad PCB design, drone frame design • Additive Manufacturing: 3D printer building, firmware tuning, prototyping • Robotics & Drones: Custom builds, flight controller programming • Systems & OS: WSL2, Linux, Windows, macOS • DevOps & Tooling: Git, environment debugging, dependency management TECHNICAL SKILLS Programming: Python, Rust, JavaScript, Bash, C, HTML/CSS Tools: Streamlit, VS Code, Android Studio, STM32 Cube Programmer, KiCad, Fusion 360 Hardware: ESC calibration, PCB prototyping, microcontroller programming AI Tools: ONNX Runtime, TTS pipelines Operating Systems: Windows, Linux, WSL2, macOS ENGINEERING EXPERIENCE Re-Hardwire – Founder/Engineer • Built modular AI-powered emotional recovery application using Python and Streamlit • Implemented ONNX-based TTS models and multi-modal AI systems • Developed routing engine, state management, and modular architecture • Managed WSL2 rebuilds, dependency conflicts, and server debugging Custom Drone Engineering Projects • Built custom drones including frame design, ESC configuration, sensor integration • Programmed flight controllers and tuned flight characteristics • Designed wiring harnesses and PCB components • Performed serial debugging and microcontroller configuration 3D Printing & Additive Manufacturing • Built and modified 3D printers with custom firmware • Designed 3D printable components using CAD tools • Calibrated printers and optimized print quality Embedded & Electrical Engineering • Designed PCBs using KiCad • Programmed STM32 microcontrollers • Built and tested circuits and performed soldering Software Development & Web Projects • Developed automation tools and scripts • Built beginner-level Rust applications • Created websites using HTML, CSS, JavaScript, Python EDUCATION & TRAINING MMGD AVP Program – Multimedia, Game Development, Audio/Visual Production Self-directed engineering training across software, hardware, robotics, and AI systems ADDITIONAL EXPERIENCE • AI pipeline development integrating SAM2, GPT-J, LLAMA • Industrial CNC and 30kW laser cutter operation • Warehouse systems, POS upgrades, SIMS, ITAR/EMAC compliance • OSHA safety practices and forklift certification